

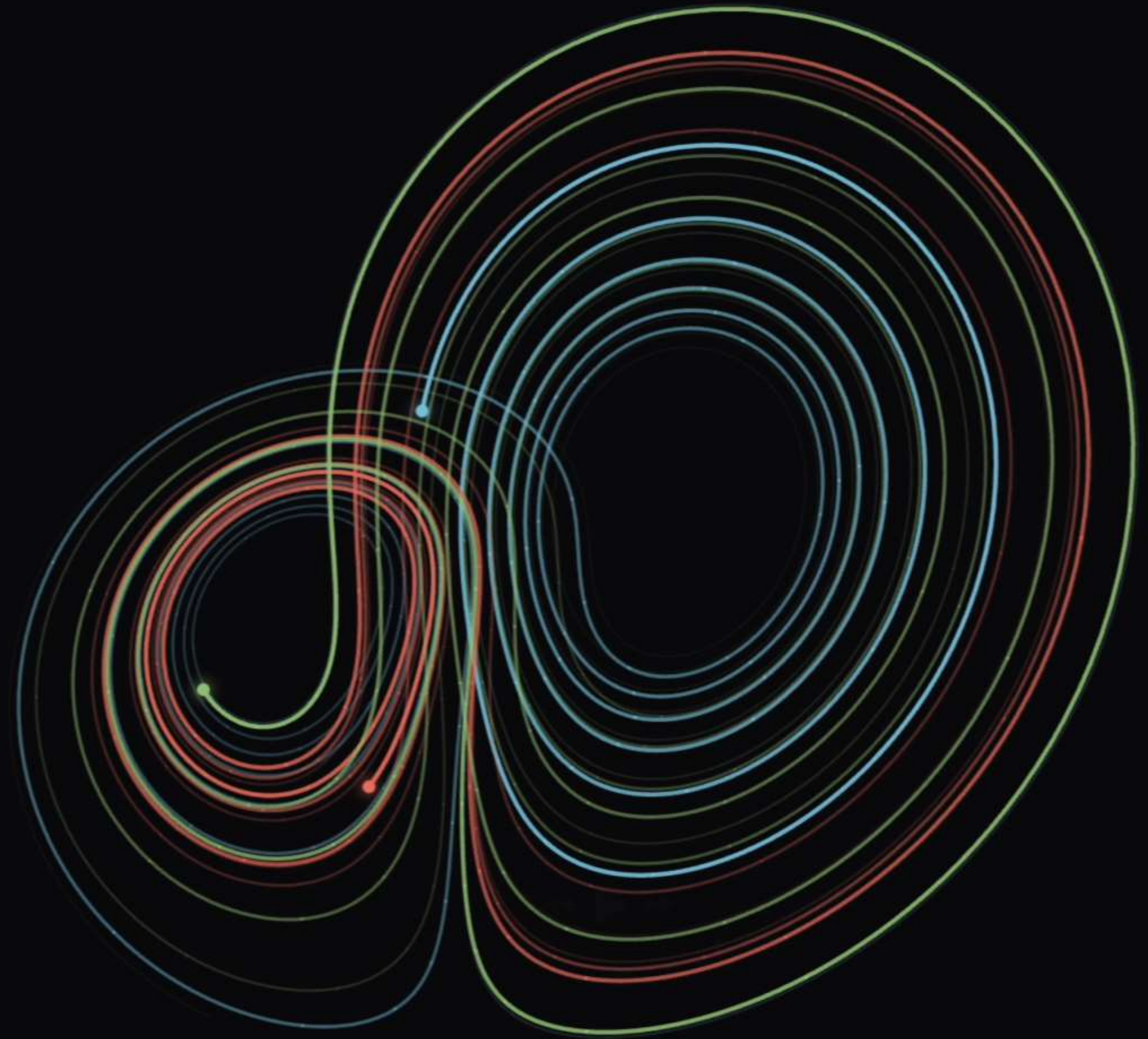
ICARM Summer School on Formalization:

Formalizing the Trapping Region Lemma for the Lorenz System in Lean

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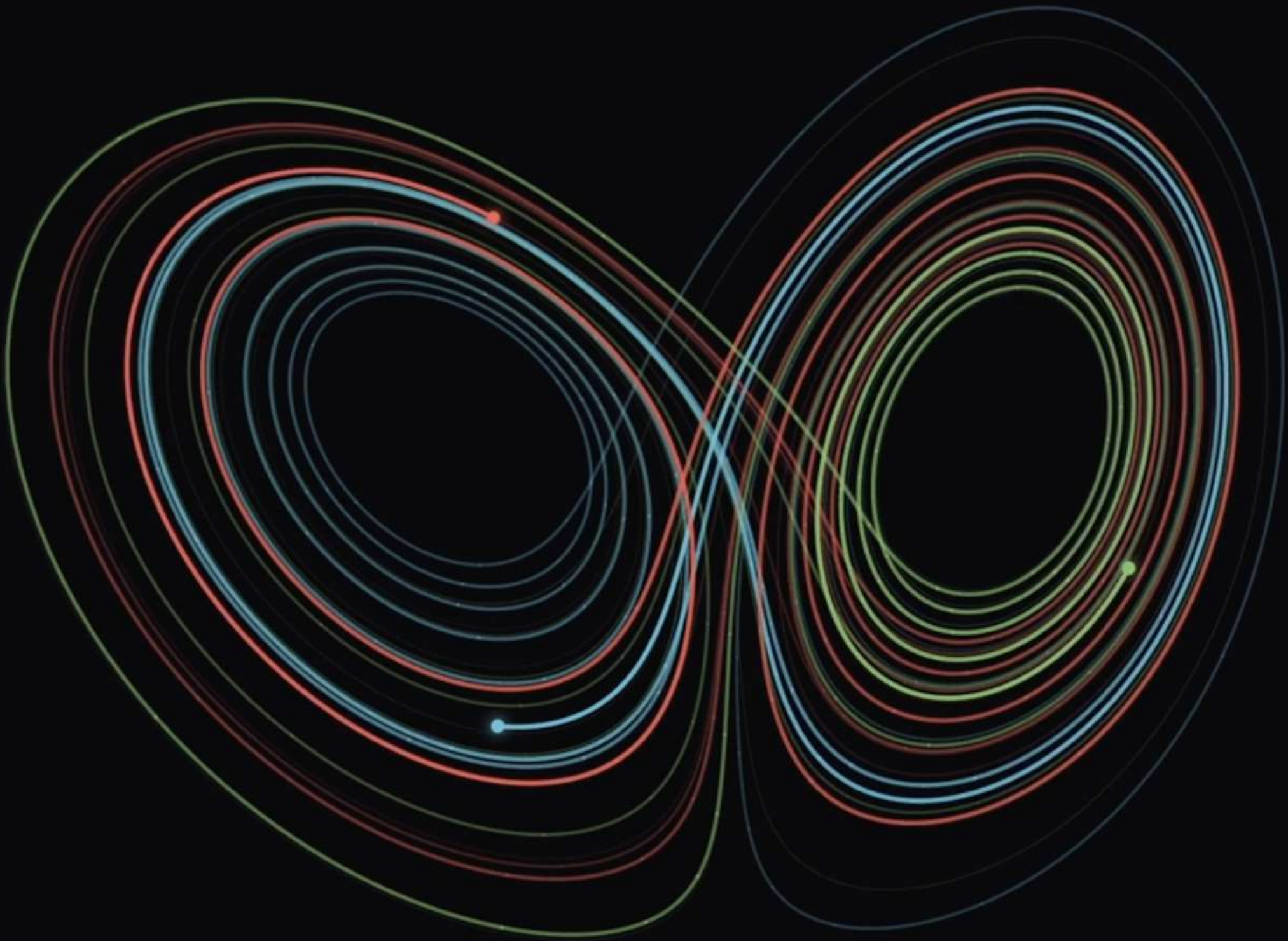


The Lorenz system of equations

A set of three coupled, nonlinear ordinary differential equations initially developed by MIT meteorologist Edward Lorenz in 1963.

$$\begin{aligned}\frac{dx}{dt} &= \sigma (y - x) \\ \frac{dy}{dt} &= x (r - z) - y \\ \frac{dz}{dt} &= xy - bz\end{aligned}$$

- $x_0 = 10.00000000$
- $x_0 = 10.00001000$
- $x_0 = 10.00002000$



TIME
46.50

SEPARATION
27.736

CHAOS THRESHOLD r_H
24.74

r is above it — chaotic

Pause

Restart

INITIAL GAP

10⁻⁵

SPEED

0.8x

r

28.0

σ

10.0

b

2.67

TRAIL

long

Trapping Lemma for the Lorenz System

Let

$$\begin{aligned}\dot{x} &= \sigma(y - x), \\ \dot{y} &= x(r - z) - y, \\ \dot{z} &= xy - bz.\end{aligned}$$

Lemma. *There exists a bounded ellipsoid E that is a trapping region for the Lorenz flow. Equivalently, the time-one map T of the Lorenz flow satisfies*

$$T(E) \subset \text{int}(E).$$

Sketch of the Proof

- ① Lyapunov function. Define

$$V(x, y, z) = rx^2 + \sigma y^2 + \sigma(z - 2r)^2.$$

Along solutions of the Lorenz system,

$$\dot{V} = -2\sigma(rx^2 + y^2 + bz^2 - 2brz).$$

- ② Negativity outside a large ellipsoid.

The quadratic form $Q(x, y, z) = rx^2 + y^2 + bz^2 - 2brz$ is positive definite away from a bounded region. Therefore

$$\dot{V} < 0 \quad \text{outside a sufficiently large ellipsoid.}$$

- ③ Construct the sets D and E .

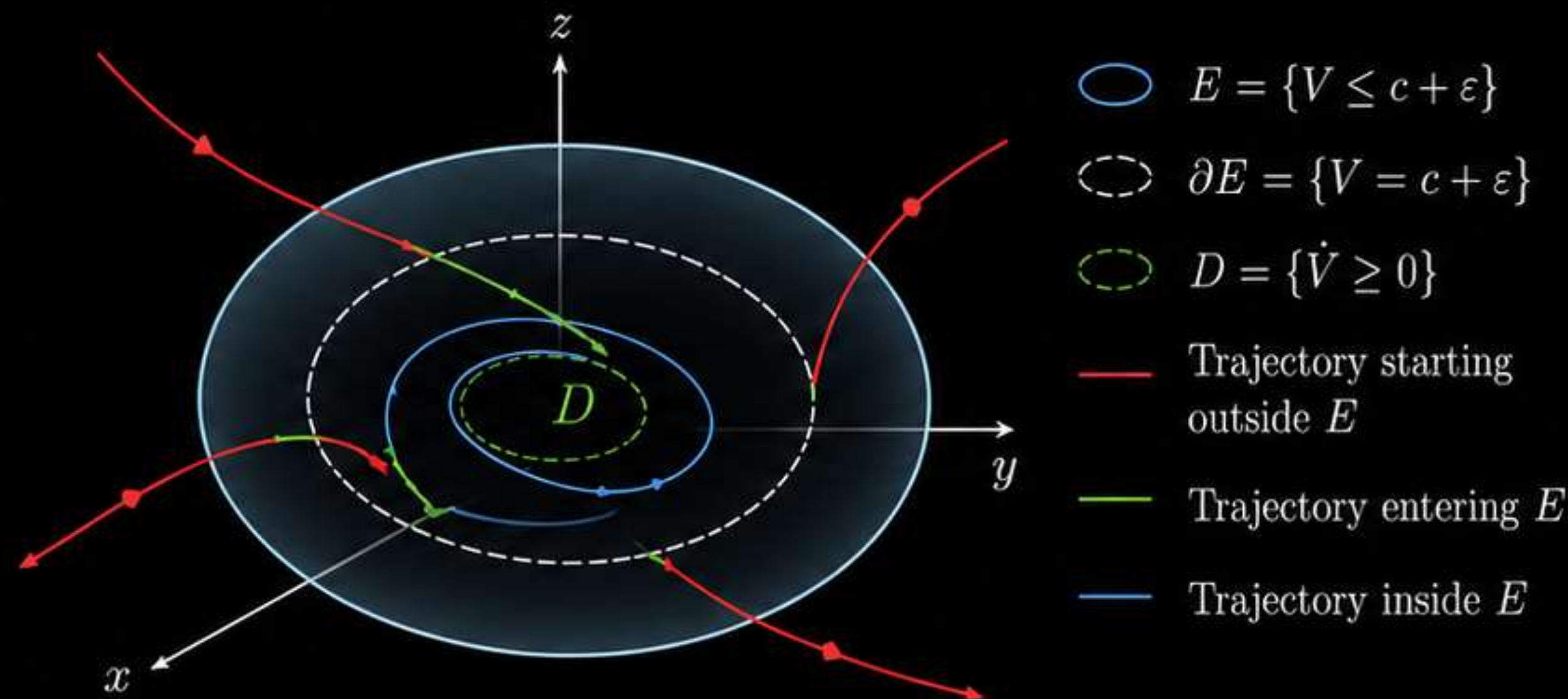
Let $D = \{\dot{V} \geq 0\}$. Then D is bounded (hence compact).

Let $c = \max_D V$ and $E = \{V \leq c + \varepsilon\}$

for some $\varepsilon > 0$. Then E is a bounded ellipsoid containing D .

- ④ Properties of E .

- **Outside E :** $\dot{V} < 0$. Hence $V(t)$ strictly decreases along any trajectory that starts outside E .
- Thus every trajectory enters E in finite time.
- On the boundary $\partial E = \{V = c + \varepsilon\}$ we have $\dot{V} < 0$, so the vector field points strictly inward. Therefore trajectories **cannot leave** E once inside.



$\Rightarrow$ Therefore E is a trapping region for the Lorenz flow. Equivalently, $T(E) \subset \text{int}(E)$.

- Phase 1 · The dynamical system
- Phase 2 · Defining V and $\tilde{V}$
- Phase 3 · The set D and $\sup c$
- Phase 4 · The set E and ∂E
- Toolkit · replaces Case 1 subcases 1–2
- Phase 5 · Trapping region

STEP 31 OF 31 · THEOREM · PHASE 5 · THEOREM

Lorenz_trapping_region

"If $\phi(t_0) \in E$, then $\phi(t) \in \text{int}(E)$ for all $t > t_0$." Four moves: use $F \text{ p}$ to supply the witness, $F_isCompact.isBounded$ to discharge the boundedness conjunct, $F_{equalBdunionInt}$ to split F into interior $\cup$ frontier, and then each piece to its case. Note that the conclusion is **stronger than the hypothesis**: land anywhere in F — even exactly on the boundary — and at every later instant you are strictly inside. The trapping region does not merely hold you; it pulls you off its own edge.

```

theorem Lorenz_trapping_region (p) (φ)
  (hφ : IsSolution p φ) :
  ∃ F : Set (ℝ × ℝ × ℝ),
    Bornology.IsBounded F ∧
    ∀ t₀ : ℝ, φ.total t₀ ∈ F →
      (∀ t > t₀, φ.total t ∈ interior F) := by
  use F p
  rw [and_iff_left ?_]
  · refine IsCompact.isBounded ?_
    apply F_isCompact
  · intro t₀ h1

```


Next Steps

① Contribute the boundary lemmas to Mathlib.

Goal. Generalize and upstream the auxiliary topological results developed for the trapping region proof.

- `firstExit_mem_frontier` *The first exit time lies on the frontier of the set.*
- `firstExit_mem_closure` *A right-continuous flow maps the first exit time into the closure of the complement.*

These lemmas are completely independent of the Lorenz system and are potentially useful in broader formalizations.

② Extend the formalization of the Lorenz system.

Prove additional fundamental results, including

- Global Existence Theorem
- Existence of an Attracting Set
- Further qualitative properties of the Lorenz flow.

③ Generalize the trapping region lemma.

Develop a version applicable to more general dynamical systems, including

- Higher-dimensional systems
- Non-smooth flows
- Infinite-dimensional systems

⇒ Towards a reusable formal library for trapping regions and dynamical systems in Lean.